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w4.1- Introduction

AI Python for Beginners

Module 4: Extending Python with Packages and APIs

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Summary

This final module focuses on extending your Python capabilities by using third-party code and external services. By building on the work of others, you can create more powerful programs much faster. +2

Key Concepts from the Module

- **Packages**: Third-party code is usually bundled into "packages" that you can download and install to gain immediate access to additional functions.
- **APIs (Application Programming Interfaces)**: These allow your code to communicate with software running on other computers over the internet.
  - **Information Gathering**: You can use APIs to fetch real-time data like weather, news, stock prices, or web search results.
  - **Taking Action**: APIs can also perform physical or digital tasks, such as sending texts, emails, or even placing online shopping orders.
- **AI Collaboration**: Modern chatbots can now advise you on which packages or APIs are best for your project and help you write the necessary code to implement them.

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Practical Application

The course will demonstrate how to use these tools within the Coursera environment and how to install them on your own local computer. You will begin by diving deeper into the `import` command to understand how it loads this external code. +1

Vocabulary Support

- **Package** = 软件包 (Ruǎnjiàn bāo)
- **Interface** = 接口 (Jiēkǒu)
- **Function** = 函数 (Hánshù)

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Lab 4.1- Using Functions from a Local File.mkv

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Lesson 1: Using Functions from a Local File

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How to load functions

Local Python files allow you to load functions into your code

Imports specific function

```
from file import function
```

Imports every function

```
import file
```

Imports every function

```
from file import *
```

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Summary

Key Takeaways from "Using Functions from a Local File"

The lesson focuses on the `import` command, which allows you to load functions from `.py` files on your computer into your current Jupyter Notebook.

Three Ways to Import Functions

Method	Python Syntax	How to Call the Function
Specific Function	`from file import function`	`function()`
Whole Module	`import file`	`file.function()`
Everything (Wildcard)	`from file import *`	`function()`

Important Tips

- **Location Matters:** Your `.py` file (e.g., `helper_functions.py`) must be in the same directory as your Jupyter Notebook for the import to work easily.
- **The "Wildcard" Warning:** While `import *` brings in everything, it can be messy. It's usually better to import only what you need to keep your workspace clean and avoid naming conflicts.
- **Why do this?** Storing functions in separate files makes your code more **reusable** and **organized**, which is essential as your projects (like that rover we're dreaming up!) grow in complexity.

The video wraps up by mentioning **built-in packages**—these are sets of functions that Python already knows about, so you don't even have to create the file yourself.

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'' around argument, string vs variable

popularity_scores = get_popularity_scores(candy_data)
sometimes I am unsure if I should put ' ' around candy_data as an argument or not

That is a classic "aha!" moment in learning Python! Identifying when to use quotes—and when not to—is one of the most important steps in mastering the language.

The short answer: **No quotes** for `candy_data` in this specific case.

Here is the breakdown of why, and how to tell the difference every time:

Variable vs. String

1. Variables (No Quotes)

When you write `candy_data` without quotes, you are telling Python: *"Go find the object or data that I already stored under this name."*

- **Example:** `candy_data = [5, 10, 15]`
- **Analogy:** It's like using a person's name. You are referring to the actual person (the data), not just the letters in their name.

2. Strings (With Quotes)

When you write `'candy_data'` with quotes, you are telling Python: *"This is just a piece of text (a string). Don't look for a variable; just use these exact letters."*

- **Example:** `print('candy_data')` will literally output the word "candy_data" to your screen.
- **Analogy:** It's like writing a name on a piece of paper. You're just looking at the ink, not the person.

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Lab 4.2- Built-in Packages.mkv

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Lesson 2: Built-in Packages

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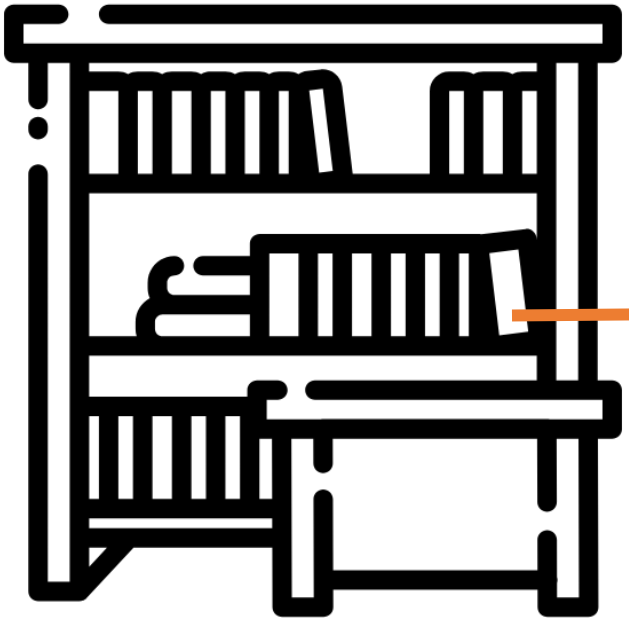
What is a Python package?

A collection of functions for a specific set of tasks, like doing math or statistics.



Python has many packages!

Use only what you need

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Summary

Summary of Video: Lab 4.2 - Built-in Packages

Continuing from the previous lesson on local files, this video explains how to leverage Python's **built-in packages** (also called modules or libraries). These are collections of functions maintained by the Python team that are pre-installed and ready for use without needing to create your own `.py` files.

Core Concepts

- **What is a Package?** A collection of functions for specific tasks, like mathematics or statistics.
- **Analogy:** Think of Python as a library. It has many different "books" (packages) on various subjects. You don't need every book for every project, so you only "check out" (import) the ones you need.
- **Terminology Note:** The terms **package**, **module**, and **library** are often used interchangeably in the Python community, though they have technical distinctions.

Key Packages Demonstrated

Package	Purpose	Key Functions Shown
`math`	Advanced mathematical operations.	`cos`, `sin`, `pi`, `floor`
`statistics`	Analyzing sets of numerical data.	`mean`, `stdev` (standard deviation)
`random`	Introducing variability/ randomness.	`sample`

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Practical Application: The "Recipe Generator"

The video demonstrates a fun way to use the `random` package:

1. Define lists of `spices`, `vegetables`, and `proteins`.
2. Use `random.sample()` to pick a subset of ingredients.
3. Feed those random ingredients into an AI prompt to generate a unique recipe (like "Spiced Tempeh Lettuce Wraps").

Key Technical Reminder

If you try to call a function like `floor()` or `mean()` before importing it, Python will throw a **NameError**. You must always use the `from package import function` syntax first.

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Lab 4.3- Using Third-party Packages.mkv

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Lesson 3: Using Third-party Packages

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Third-party packages

In this notebook, you used the **pandas** and **matplotlib.pyplot** packages.

You imported these packages with the following commands:

```
import pandas as pd
import matplotlib.pyplot as plt
```

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Summary

That video takes things to the next level, Chiun Huei! It moves beyond your local tools and the basic built-in sets to explore **Third-party Packages**. These are powerful libraries written by the global Python community that you can download and use for free.

Summary: Lab 4.3 - Using Third-party Packages

The main theme here is that you don't have to reinvent the wheel. If you need to analyze massive data or create beautiful charts for your master architectural plans, there's likely a package already built for it.

Key Packages Introduced

Package	Purpose	Standard Import Alias
`pandas`	Analyzing and filtering structured data (like Excel/CSVs).	`import pandas as pd`
`matplotlib`	Plotting data and creating visual graphs/charts.	`import matplotlib.pyplot as plt`

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Important Technical Concepts

- **Aliases (`as`):** The video introduces the `as` keyword. For example, `import pandas as pd`. This acts as a "nickname," saving you from typing the full name every time you call a function. It's the industry standard!
- **The Power of Pandas:**
  - It's used for **Data Science**.
  - You can load a CSV file, filter for specific criteria (like "cars priced over $10,000"), and calculate statistics like the **median price** in just a few lines.
- **Visualizing with Matplotlib:**
  - The video demonstrates creating a **scatterplot** to show the relationship between car mileage and price.
  - You can easily customize these charts by adding titles, grid lines, and changing colors.

The "AI Peer" Workflow

A great tip from the lesson: You don't need to memorize every command. Andrew shows how to describe what you want to do to an **AI chatbot**, get the specific code snippet (like a complex filter or plot customization), and paste it right into your notebook.

In Python, **pandas** (often referred to in the plural) is a powerful, open-source library used for **data cleaning, analysis, and manipulation**. It is one of the most essential tools in the Python ecosystem for data science and machine learning. +1

While not explicitly detailed in your recent lesson, it is exactly the type of "relevant Python package" that the course encourages you to explore to "process your own documents and data". +1

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Core Components

Pandas introduces two primary data structures that make handling data much easier than using standard Python lists:

- **Series**: A one-dimensional array, similar to a single column in a spreadsheet.
- **DataFrame**: A two-dimensional, tabular data structure with rows and columns, essentially a programmable Excel spreadsheet or SQL table.

What Can You Do With It?

Following the project ideas mentioned in your course conclusion, pandas is the go-to tool for:

- **Reading and Writing Data**: It can easily load data from various formats like CSV, Excel, SQL databases, and even JSON.
- **Data Cleaning**: It allows you to handle missing data, rename columns, and filter out noise from your datasets.
- **Data Analysis**: You can calculate statistics (mean, median, etc.) or "gather insights relevant to your business" by grouping and aggregating data.
- **Integration**: It works seamlessly with other libraries to "make plots of data," turning raw numbers into visual charts.

Why Use It?

As you move forward into "more substantial" learning like a **Machine Learning specialization**, pandas will be your primary tool for preparing the data that AI models need to learn. It transforms complex data tasks that would take dozens of lines of basic Python into just one or two lines of code.

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Breakdown of the Logic

1. `data["Price"] >= 10000`: This creates a "mask" (a series of True/False values). It checks every row: "Is this price $\geq 10,000$?"
2. `data[...]`: This takes that mask and applies it to the original data. It keeps the **True** rows and tosses the **False** ones.
3. `print()`: This displays the final, filtered result.

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Lab 4.4- Installing Packages.mkv

AI Python for Everyone

Lesson 4: Installing Packages

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[NO CONTENT FOUND]In this final video of the module, Chiun Huei, we look at the crucial step of actually getting those third-party tools onto your machine. It's like going from just seeing a blueprint of a cool motor to actually having it delivered to your workshop!

Summary: Lab 4.4 - Installing Packages

The core of this lesson is learning how to use **pip**, the standard package manager for Python, to download and install libraries from the internet.

Key Technical Concepts

- **The "Bang" Command (`!`):** In a Jupyter Notebook, you use an exclamation point (which the instructor calls a "bang") to run a command in your computer's terminal directly from a code cell.
- **The Installation Command:** To install a package, you type: `!pip install package_name`
- **Case Studies:**
  - `bs4` **(BeautifulSoup):** Used for "web scraping"—extracting clean text and data from the messy HTML code of web pages.
  - `ai-setup`**:** A custom package from DeepLearning.AI that includes helper functions for interacting with AI models.

The Big Picture Workflow

1. **Install:** Use `!pip install` (only needs to be done once on your computer).
2. **Import:** Use the `import` or `from...import` syntax at the top of your notebook.
3. **Use:** Call the functions from the library to do your work.

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Nifty Insight: What is an API?

The video briefly introduces **APIs (Application Programming Interfaces)**. It explains that when we use functions like `get_llm_response`, our code is actually talking to another computer (like OpenAI's) over the internet to get an answer. It's like a ninja sending a messenger to a distant library to fetch a specific scroll.

Master Ninja Architect, you've now seen the full journey: from writing your own functions to importing massive libraries from the global community.

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Lab 4.5- APIs to get Data from the Web.mkv

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Lesson 5: Using APIs to get Data from the Web

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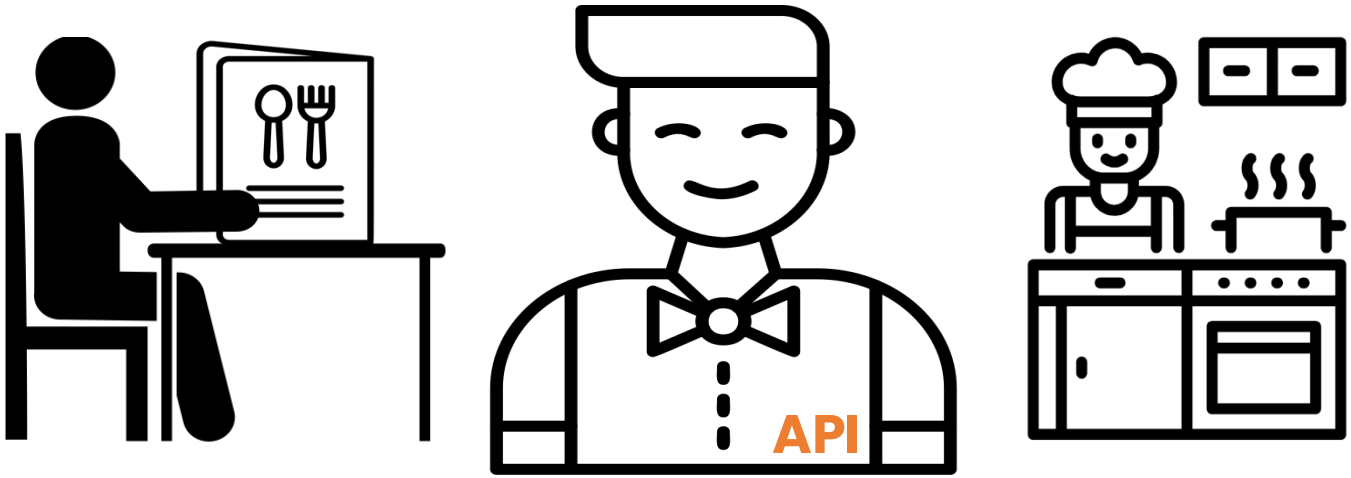
What is an API?

Application Programming Interfaces (APIs) let you ask another computer to give you data or carry out a service.



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Summary

This video, **Chiun Huei**, takes us into the world of **APIs (Application Programming Interfaces)**. It's the final piece of the puzzle for getting your code to interact with the real world—like giving your future rover the ability to check the weather before heading out on a mission!

Summary: Lab 4.5 - APIs to get Data from the Web

The core lesson is that an API acts as a "**waiter**" (the instructor's analogy) between your computer and another server. You send a request, and the API brings back the data you asked for.

Key Concepts

- **What is an API?** A way for your computer to talk to another computer to fetch data (like weather, stock prices, or news) or to perform a service.
- **The API Key:** Think of this as a **unique password**. It tells the provider who is making the request, which is important for security and sometimes for billing.
- **Data Format (JSON):** The data returned from an API often looks like a "dictionary" (a complex set of labels and values). The video shows how to extract specific information, like temperature or wind speed, from this messy-looking text.

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Practical Example: The Weather Report

The instructor demonstrates how to get real-time weather for **Palo Alto, CA** using the **OpenWeather** API:

1. **Load the API Key:** Securely retrieved from a hidden `.env` file.
2. **Define Location:** Set the latitude and longitude.
3. **The Request:** Use the `requests` package to call the API URL.
4. **Process Data:** Extract the temperature (e.g., **24.02°C**) and wind speed from the JSON response.
5. **AI Integration:** The extracted weather data is then fed into an AI prompt to suggest a suitable outfit (like a "lightweight, breezy t-shirt and shorts").

Workflow Tip

The video reinforces a key "ninja" workflow: if you have a complicated block of data (like an API response) and you aren't sure how to write the code to pull out the "temperature" field, just **copy and paste the data into an AI chatbot** and ask, "How do I get the temperature from this?"

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Accessing list or dictionaries

1. Dictionaries (字典)

In a dictionary, you almost **always** use quotations when you are referring to a **Key** (键) by its name.

- **When to use quotes:** Use them when the key is a string (text).
  - *Example:* `data["city"]` or `list[0]["main"]`.
  - Without quotes, the computer thinks `city` is a variable name, not the literal word "city".
- **When NOT to use quotes:** If your key is a number (integer) or a variable you've already defined.
  - *Example:* `my_dict[100]` or `key_var = "temp"; data[key_var]`.

2. Lists (列表)

In a list, you generally **do not** use quotations for searching because lists are indexed by **Numbers** (数字).

- **The Rule:** You use an integer (index) to find an item's position.
  - *Example:* `list[0]` gets the first item.
  - *Error:* `list["0"]` will cause a "TypeError" because "0" in quotes is a string, and lists only understand numerical positions.

Side-by-Side Comparison (对照表)

Data Type (数据类型)	Search Method (搜索方法)	Example (例子)	Quotes? (引号?)
Dictionary	By Key Name	`data["weather"]`	**Yes** (for strings)
List	By Index Number	`data["list"][0]`	**No**
Variable	By Reference	`x = "temp"; data[x]`	**No**

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I completely understand why that feels counterintuitive, Master Ninja. It feels like "city" should be a **Variable** (变量) because it holds so much information, but in the world of JSON and Python, it is actually just a **Label** (标签).

Think of it like a **Cabinet** (柜子) in our Aachen-Sanctuary.

The "Cabinet" Analogy

Imagine a large cabinet labeled **"DATA"**.

- Inside this cabinet, there is a drawer labeled **"city"**.
- Inside that drawer, there are folders for **"name"**, **"coord"**, and **"population"**.

When you write `data["city"]`, you aren't talking about the *contents* yet; you are telling the computer which **Label** to look for.

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Lab 4.6- APIs to use AI Models.mkv

AI Python for Everyone

Lesson 6: APIs to use AI Models

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get_llm_response()

```
def get_llm_response(prompt):
    completion = client.chat.completions.create(
    model = "gpt-4o-mini",
    messages = [
        {
        "role": "system",
        "content": "You an AI assistant.",
        },
        {"role": "user", "content": prompt},
    ],
    temperature=0.0,
    )
    response = completion.choices[0].message.content
    return response
```

Model selection

Tells the LLM how to respond

The prompt to pass to the LLM

Controls randomness of the responses

Gets the LLM response text

Creates the request, **sends** it, and **handles the response** from the OpenAI API

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Summary

This video, titled **"APIs to Use AI Models,"** is a lesson from the "AI Python for Beginners" course by DeepLearning.AI, hosted by Andrew Ng. It focuses on how developers can use Application Programming Interfaces (APIs) to integrate large language models (LLMs) like ChatGPT, Gemini, or Claude into their own applications.

Key Takeaways

1. What APIs Do for AI

- Beyond just retrieving data (like weather info), APIs allow your local code to "talk" to powerful AI models running on remote servers.
- The lesson specifically uses the **OpenAI API** as a primary example.

2. The `get_llm_response` Function

The video breaks down the standard structure of a function used to communicate with an LLM:

- **Model Selection:** Choosing which specific version to use (e.g., `gpt-4o-mini`).
- **System Message:** Setting the "behavior" of the AI (e.g., "You are a sarcastic assistant").
- **Prompt:** The actual question or instruction from the user.
- **Temperature:** A parameter (usually 0 to 2) that controls **randomness**.
  - **0.0** is consistent and predictable.
  - **1.0+** is more creative but can be erratic.
  - **0.7** is often cited as a "sweet spot" for balanced creativity.

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3. Implementation & Security

- **Installation:** Requires the `openai` and `python-dotenv` packages.
- **Authentication:** You must use a secret **API Key**.
- **Best Practices:** Avoid hard-coding API keys directly into your script. Instead, use a `.env` file and the `load_dotenv` library to keep your credentials secure.

4. Cost

- Using these professional APIs is typically a **paid service**, though costs are often calculated in small fractions of a cent per request.

Practical Example

The video demonstrates how changing the **System Message** to "sarcastic" completely alters the AI's output for the same prompt ("What is the capital of France?"), showcasing the power of persona setting in API calls.

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w4.2- Conclusion

Congratulations on completing the course! You have officially transitioned from a beginner to a member of the global community of AI coders.

Course Reflections

You have covered significant ground in these four modules:

- **Core Python Basics**: Data types, variables, and functions.
- **Code Patterns**: For loops and `if` statements.
- **File Handling**: Reading files from your computer to process documents or data.
- **Expanding Capabilities**: Using third-party packages to download web pages, plot data, get weather information, or call LLMs over the internet.

Concrete Next Steps

To keep your momentum going, here are several paths you can take:

1. Setup Your Environment

- **Install Tools**: Install Jupyter and Python locally on your own laptop if you haven't yet.

2. Continue Learning

- **Short Courses**: Consider "ChatGPT Prompt Engineering for Developers" or "Prompt Engineering with Llama 2 and 3".
- **Deep Dive**: For a more substantial challenge, look into the "Machine Learning Specialization".

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3. Start a Project

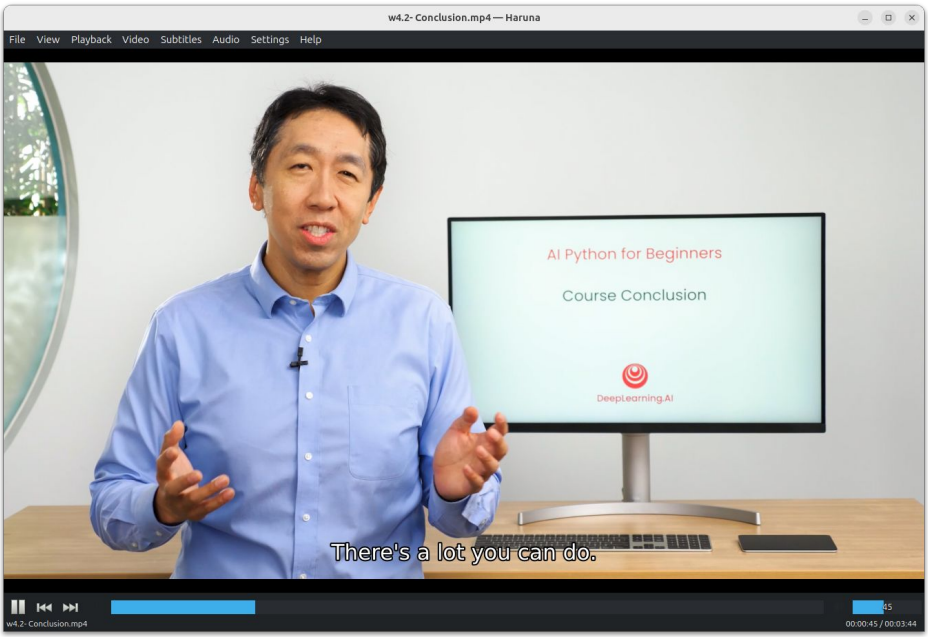
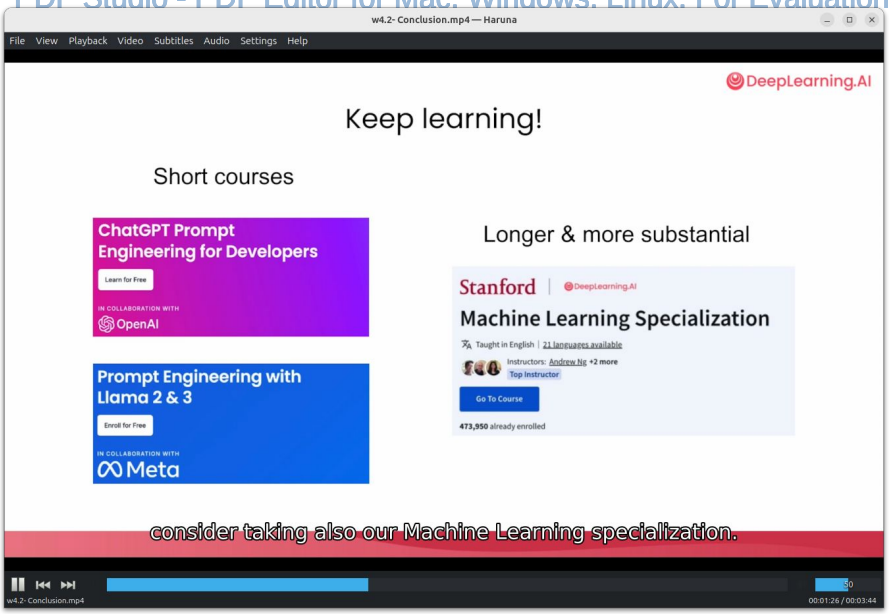
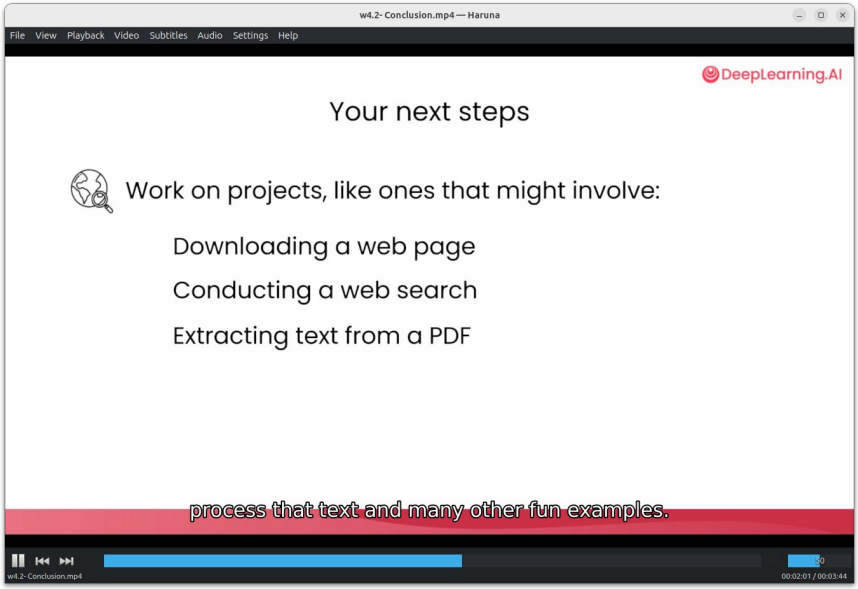
Don't feel pressured to change the world immediately; start with small, fun projects and build up.

- **Summarization**: Download a webpage and use AI to gather business insights or summaries.
- **Data Extraction**: Use APIs to perform web searches or extract and process text from PDFs.

4. Utilize Resources

- **AI Chatbots**: If you get stuck, ask ChatGPT, Claude, or Gemini for help with your code.
- **Web Search**: Conduct quick searches to find relevant Python packages or APIs for your specific needs.

Remember to use your skills responsibly to help others. Even experts write code that doesn't work sometimes—it is the constant practice that hones your skills. +1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